

Eye Contact and Engagement Analysis Report

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Section: BCS-7A

1. Approach

The project aims to classify gaze engagement into three categories attentive, disengaged, and distracted using image-based cues from the eye region. The approach focuses on precise preprocessing, an efficient MobileNet model, and a training strategy that handles imbalance and improves generalization.

2. Methodology

2.1 Dataset Analysis and Label Extraction

Images contain pitch (P), vertical (V), and horizontal (H) gaze angles in their filenames. A custom regex-based extractor parsed these values to understand dataset pose distribution.

Label analysis showed:

- Attentive: 9 images
- Disengaged: 42 images
- Distracted: 54 images

2.2 Preprocessing Pipeline

Eye-region cropping was applied using a fixed-region approach to isolate gaze-relevant features. Brightness and contrast normalization techniques (histogram equalization, contrast stretching, and normalization) ensured visual consistency. All images were resized to 224×224 and scaled before training.

2.3 MobileNet Architecture

MobileNet (include_top=False) was used for its lightweight and efficient structure. A custom classification head included:

- Global Average Pooling
- Dense (ReLU)
- Dropout
- Softmax layer for 3 classes

The model used Adam optimizer and categorical cross-entropy loss.

2.4 Training Strategy

Stratified K-Fold (5 splits) ensured balanced class proportions across folds. Class weights were applied to address imbalance, giving the attentive class the highest weight. EarlyStopping and ModelCheckpoint callbacks were used for stability.

3. Initial Results

3.1 Class Weights

Example:

{0: 4.67, 1: 1.00, 2: 0.78}

This improved minority-class learning.

3.2 Performance Metrics

Accuracy: 89%

Precision: 81%

Recall: 93%

F1 Score: 83%

3.3 Training Behavior

Training logs showed:

- Faster convergence with MobileNet
- Improved validation accuracy across folds
- Better recall for the attentive class
- More stable learning after cropping, normalization, and weighting

3.5 Result Summary

The pipeline showed strong stability, effective handling of imbalance, and improved feature clarity from preprocessing. MobileNet performed efficiently despite the limited dataset size.